



# Experiment-3

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**Semester:** 5th  
**Subject Name:** Computer Networks

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**Date:** 29/07/2026  
**Subject Code:** 24CSH-337

## 1. Aim/Overview of the practical:

To simulate and understand the working of the following networking devices using Cisco Packet Tracer:

1. Hub
2. Switch
3. Router

## 2. Software/Tools Required:

- Cisco Packet Tracer
- Windows Operating System

## 3. Theory:

### A) Hub

#### What is a Hub?

A **Hub** is a basic networking device that operates at the **Physical Layer (Layer 1)** of the OSI Model. It is used to connect multiple devices in a Local Area Network (LAN). Unlike intelligent networking devices, a hub does not examine or store network addresses. Whenever a data packet is received on one port, it simply broadcasts the data to all other connected ports.

#### Characteristics of a Hub

- Operates at the **Physical Layer (Layer 1)** of the OSI Model.
- Used to connect multiple devices within a Local Area Network (LAN).
- Contains multiple ports for connecting network devices.
- Broadcasts incoming data to all connected devices.
- Does not maintain a MAC Address Table or Routing Table.
- Typically operates in **Half-Duplex** mode.
- Commonly used in small networks and for learning basic networking concepts.



### Advantages of a Hub

- Simple and easy to use.
- Less expensive than a switch.
- Suitable for small networks.

### Disadvantages of a Hub

- Broadcasts data to every connected device.
- Does not store MAC addresses.
- Generates unnecessary network traffic.
- Lower performance compared to a switch.

## B) Switch

### What is a Switch?

A **Switch** is a networking device that connects multiple devices within a Local Area Network (LAN) and enables efficient communication between them. It operates at the **Data Link Layer (Layer 2)** of the OSI Model and forwards data frames using **MAC (Media Access Control) Addresses**. Unlike a hub, a switch sends data only to the intended destination device instead of broadcasting it to all connected devices, thereby improving network performance and reducing unnecessary traffic.

### Characteristics of a Switch

1. A Switch operates at the **Data Link Layer (Layer 2)** of the OSI Model.
2. It forwards **frames** based on the destination **MAC Address**.
3. It learns and stores the MAC Address of connected devices in a **MAC Address Table (CAM Table)**.
4. It forwards data only to the required destination port, reducing network congestion.
5. Each switch port provides a separate collision domain, improving network efficiency.
6. It supports **Full-Duplex** communication, allowing simultaneous transmission and reception of data.
7. It is widely used in LANs to provide fast and reliable communication between connected devices.



### **MAC Address Table**

A Switch automatically builds a **MAC Address Table (CAM Table)** by recording the source MAC Address and the corresponding port whenever a device sends data. When another frame arrives, the switch checks this table and forwards the frame only to the correct destination port. If the destination MAC Address is unknown, the switch temporarily forwards the frame to all ports except the incoming port until it learns the correct destination.

## **C) Router**

### **What is a Router?**

A **Router** is a networking device that connects two or more different networks and forwards data packets between them. It operates at the **Network Layer (Layer 3)** of the OSI Model and makes forwarding decisions using **IP Addresses**. A router enables communication between different networks, such as connecting multiple LANs or connecting a LAN to the Internet.

### **Characteristics of a Router**

1. A Router operates at the **Network Layer (Layer 3)** of the OSI Model.
2. It forwards **packets** using destination **IP Addresses**.
3. It connects two or more different networks and enables communication between them.
4. It maintains a **Routing Table** that stores information about available networks and the best paths to reach them.
5. Each router interface belongs to a different network and is assigned a unique IP Address.
6. It separates broadcast domains, thereby reducing unnecessary network traffic.
7. It can perform both static and dynamic routing depending on the network configuration.

## Routing Table

A **Routing Table** is a database maintained by the router that contains information about directly connected and remote networks. Whenever a packet arrives, the router checks the destination IP Address against the routing table and forwards the packet through the most appropriate interface. This process ensures that data reaches the correct destination efficiently.

## Default Gateway

A **Default Gateway** is the IP Address of the router interface that allows devices in one network to communicate with devices in another network. When a destination IP Address belongs to a different network, the host forwards the packet to the default gateway, and the router routes it to the appropriate destination network.

| Feature                   | Hub                            | Switch                                   | Router                                 |
|---------------------------|--------------------------------|------------------------------------------|----------------------------------------|
| <b>OSI Layer</b>          | Layer1<br>(Physical)           | Layer 2 (Data Link)                      | Layer 3 (Network)                      |
| <b>Data Unit</b>          | Bits                           | Frames                                   | Packets                                |
| <b>Address Used</b>       | No Addressing                  | MAC Address                              | IP Address                             |
| <b>Data Transmission</b>  | Broadcasts to all devices      | Sends data to the destination device     | Routes data between different networks |
| <b>Table Maintained</b>   | None                           | MAC Address Table                        | Routing Table                          |
| <b>Communication Mode</b> | Half-Duplex                    | Full-Duplex                              | Full-Duplex                            |
| <b>Primary Use</b>        | Connect devices in a small LAN | Connect devices efficiently within a LAN | Connect different networks (LAN/WAN)   |

#### **4. Steps for Experiment/Practical:**

##### **A) Hub**

Procedure / Steps:

1. Open Cisco Packet Tracer.
2. Place one Hub and four PCs in the workspace.
3. Connect each PC to the Hub using Copper Straight-Through cables.
4. Assign IP addresses to all the PCs in the same network.
5. Open the Command Prompt on one PC.
6. Execute the ping command to test connectivity with another PC.
7. Verify that the Hub broadcasts the data to all connected devices and communication is established successfully.

##### **B) Switch**

Procedure / Steps:

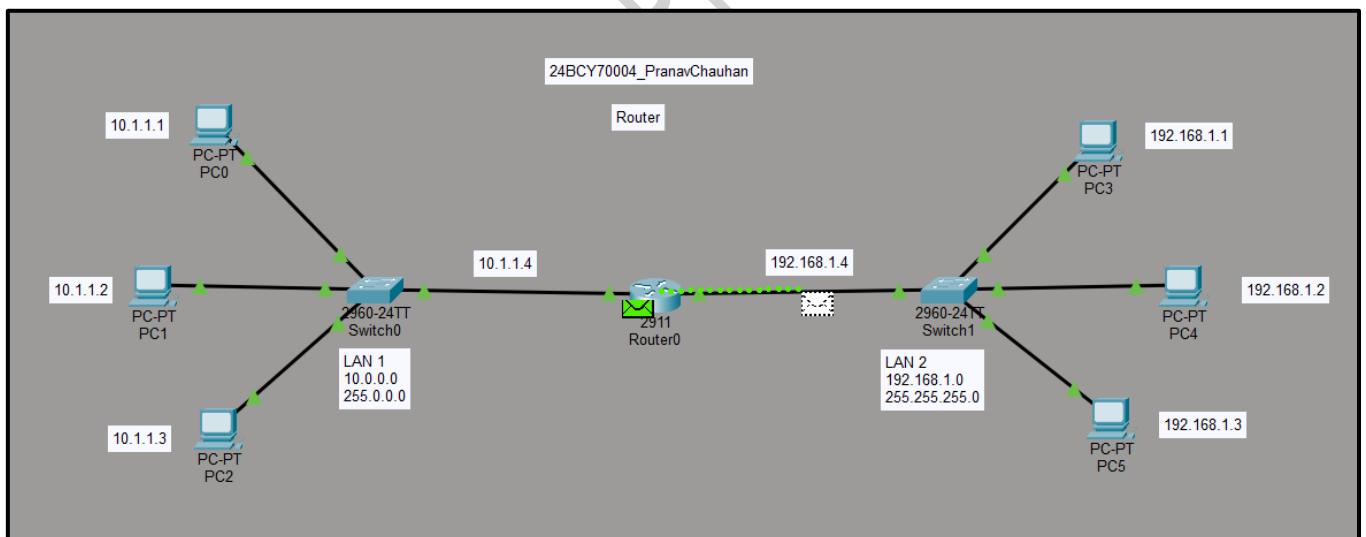
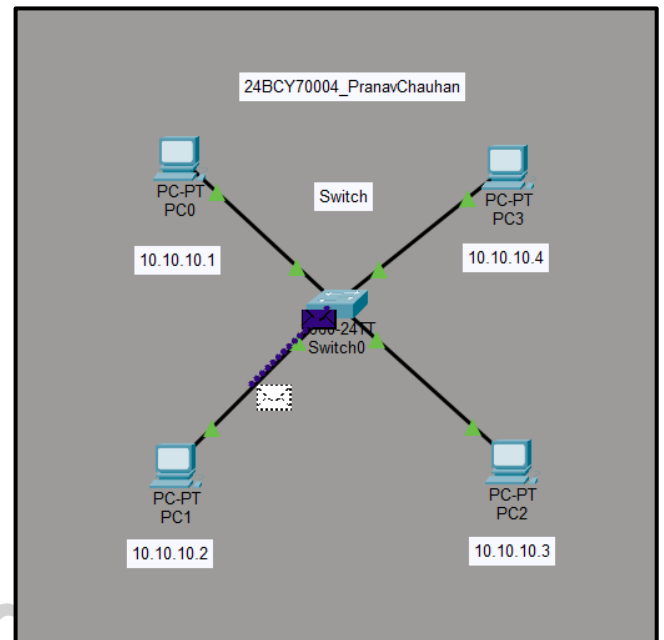
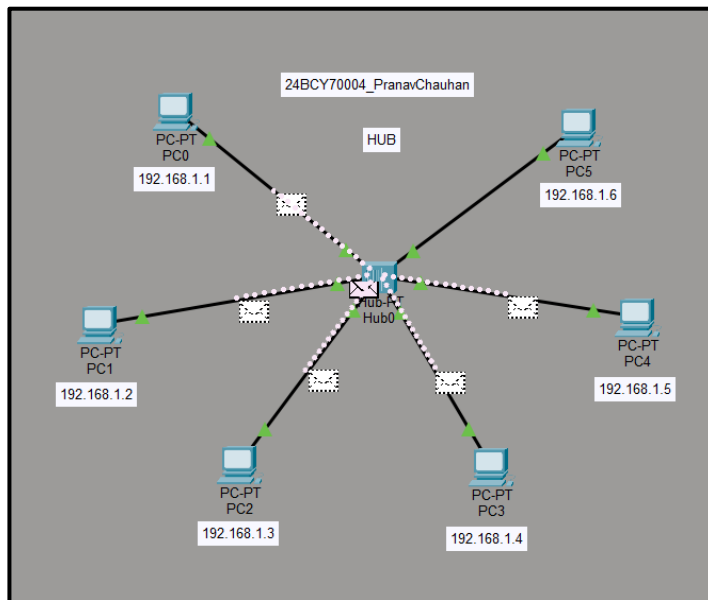
1. Open Cisco Packet Tracer.
2. Place one Switch and four PCs in the workspace.
3. Connect all the PCs to the Switch using Copper Straight-Through cables.
4. Assign IP addresses to all the PCs in the same network.
5. Open the Command Prompt on one PC.
6. Execute the ping command to communicate with another PC.
7. Verify that the Switch forwards data only to the intended destination using the MAC Address Table.

##### **C) Router**

Procedure / Steps

1. Open Cisco Packet Tracer.
2. Place one Router, two Switches, and four PCs in the workspace.
3. Connect each pair of PCs to a Switch using Copper Straight-Through cables.
4. Connect both Switches to the Router using Copper Straight-Through cables.
5. Assign IP addresses to all PCs and configure IP addresses on the Router interfaces.
6. Configure the Default Gateway on each PC with the corresponding Router interface IP address.
7. Verify connectivity between the two networks using the ping command and display the routing table using:
  - show ip route

## 5. Output:



## 6. Result

The working of **Hub, Switch, and Router** was studied successfully using **Cisco Packet Tracer**. Communication between devices connected through a Hub and a Switch was verified, and successful communication between different networks was established using a Router. The functionality of data transmission, MAC address-based switching, and IP-based routing was observed and understood.

## 7. Learning Outcomes

- Understood the working and characteristics of **Hub, Switch, and Router**.
- Learned the differences between data transmission using a **Hub, Switch, and Router**.
- Configured and simulated network topologies using **Cisco Packet Tracer**.
- Verified communication between devices within the same network using a **Switch** and between different networks using a **Router**.
- Understood the concepts of **MAC Address Table, Routing Table, and IP-based communication**.